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Research Paper

Depression and related factors among public health nurses engaged in cases of child neglect-related abuse

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ABSTRACT

Objectives: To evaluate the occurrence of depression and its related factors among public health nurses (PHNs) engaged in child neglect-related abuse cases.
Methods: A cross-sectional study of PHNs involved in maternal and child health or child and family welfare, and engaged in cases of child neglect-related abuse was conducted from January to March 2023 in all municipalities of 13 prefectures selected from Japan's seven regions, considering geographic diversity and major urban areas. The Patient Health Questionnaire-2 was used to screen for depression. Statistical analyses included the Chi-square test, Mann–Whitney *U* test, and logistic regression analysis.
Results: A total of 359 provided valid responses. Among them, 11.4 % screened positive for depression. Significant individual factors related depression were management-stage nurses (*OR* = 6.27, 95 %*CI*: 1.63–24.18), no disability welfare experience (*OR* = 3.49, 95 %*CI*: 1.05–11.55), being assignment of high-risk cases within the past year (*OR* = 8.28, 95 %*CI*: 2.07–33.08), and lower work control (*OR* = 0.53, 95 % *CI*: 0.32–0.86). Workplace environment factors were lower perceptions and relationships within the workplace (*OR* = 0.43, 95 %*CI*: 0.29–0.65) and perceiving manpower as not secured (*OR* = 9.07, 95 %*CI*: 1.17–70.40).
Conclusions: Mental health measures for PHNs should include workload management and reduced burdens through workforce enhancement, attention to burdens on managerial PHNs, promoting communication and mutual support, and opportunities to engage in disability welfare work.
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What is known?

- Most public health nurses (PHNs) experience difficulties because they continuously manage child neglect-related abuse cases for long periods. They also experience psychological trauma during their work.
- The stress experienced by PHNs is expected to be high. However, few studies have focused on the mental health of PHNs engaged in child abuse prevention.

What is new?

- The prevalence of depression among PHNs engaged in child neglect-related abuse cases is 11.4 %.
- Individual factors included years of experience, experience in disability welfare work, being in charge of a case requiring cooperation with the Child Guidance Centre within the past year, and work controls. Workplace environmental factors included perceptions and relationships within the workplace and perceptions of manpower availability.
- Based on our findings, we recommended supporting the career development of PHNs and establishing healthy workplace environments.

1. Introduction

Public health nurses (PHNs) are nationally licensed nurses

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specializing in public health in Japan. Most PHNs belong to prefectural and municipal government agencies, such as public and municipal health centres. Therefore, they assist community residents, including mothers, children, adults, older adults, and people with disabilities, from a prevention standpoint. Owing to the diversity of community health issues (such as the declining birth rate and aging population) and health crises, the scope of work has broadened, and PHNs have been assigned to various departments. Municipalities provide health services familiar to residents of all ages, such as health check-ups and home visits for mothers and children, lifestyle-related disease prevention, welfare for people with disabilities, and nursing care prevention [1]. In municipalities, 32,328 PHNs are working [2]. Approximately 70 % of PHNs are assigned to health departments such as maternal and child health and adult health [3].

PHNs in the maternal and child health and child and family welfare departments of the municipality routinely provide early detection and preventive support for families at high risk of child abuse through home visits and childcare consultations [4]. In Japan, child abuse is reported to the municipality and the prefectural child guidance centre, with the city responsible for primary and secondary prevention [5]. Not only are child welfare workers responsible for addressing child abuse, but PHNs are also involved. In recent years, an increase in the number of child abuse cases has led to calls for measures to prevent outbreaks and other severe cases. Child abuse can be classified into four types: physical, psychological, neglect-related, and sexual abuse. Neglect-related abuse was the most common response among municipalities [6]. Neglect-related abuse is caused by overlapping factors, such as mental illness, disability, parental deprivation [7], and other health problems and social disadvantages, necessitating responses that consider diversity and individuality. Approximately 90 % of PHNs experience difficulties as they are continuously involved in managing numerous cases over an extended period [8]. In addition, approximately 40 % of PHNs experience psychological trauma at work when managing severely injured children or when required to make critical decisions that may affect the success or failure of their interventions, thereby significantly impacting the safety of consultants [9]. This department, which has many PHNs, has the highest number of employees on leave as a result of mental health problems [10]. According to a national survey, PHNs in the municipality spend the most time on maternal and child health and child and family welfare [11]; thus, they are assumed to have a heavy workload. Research on child abuse prevention overseas has focused on child welfare workers, many of whom have reported experiencing secondary traumatic stress while handling cases [12]. Secondary traumatic stress is reportedly related to health status, suggesting a need to improve the mental health of support workers and, in turn, client support [13]. In Japan, the stress experienced by PHNs dealing with such cases is expected to be high. The mental health of professionals who care for children can affect the care they provide [14]; consequently, mental health problems among PHNs may reduce the quality of care and service provision [15]. PHNs handling cases of neglect-related abuse experience prolonged involvement and sustained exposure to high-stress conditions [16]. PHNs may consider leaving their jobs owing to stress, burnout, and signs of perceived health problems. Despite the heavy and complex workload they face, little is known about the mental health of PHNs involved in child abuse prevention, particularly those supporting neglect-related abuse cases. This research gap is notable given that existing studies have reported significant mental health concerns among PHNs working in other high-stress departments. According to previous research, PHNs working in community general support centres have a higher frequency of depressive tendencies than the general workforce [17]. Additionally, they tend to exhibit poorer mental health perceptions and higher levels of work-related stress [18]. During the coronavirus disease (COVID-19) pandemic, approximately 80 % of PHNs experienced high levels of mental distress, and assignment to infectious disease departments affected their mental health [19]. A survey of nurses indicated that the mental health situation varies according to the department where a nurse works, as job demands and control levels differ across workplace environments [20]. Even among PHNs, the mental health situation may vary according to their department and the nature of their duties.

Therefore, this study aimed to identify PHNs who may be at risk of depression and examine the related factors. By focusing on a group that has received limited research attention, the findings of this study are expected to provide insights into the need for support for PHNs and contribute to improving their well-being and the quality of maternal and child health and child welfare services.

2. Methods

2.1. Study design and setting

This cross-sectional study was conducted from January to March 2023 and reported by the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines [21]. Based on general regional divisions, we selected 13 prefectures from all seven regions of Japan (Hokkaido/Tohoku, Chubu, Kanto, Kansai, Chugoku, Kyushu, and Shikoku): one from Shikoku, which has the smallest population and fewest prefectures, and two from each of the other six regions. Prefectures were chosen to reflect geographic and demographic diversity, including each region's largest city and major metropolitan areas. This selection was conducted using a stratified purposive sampling approach, stratifying by the seven major regions of Japan and purposively including prefectures that represented both urban and rural contexts. We targeted all 382 municipalities within these prefectures, covering 534 maternal and child health departments (366 in 214 cities and 168 in towns and villages).

2.2. Study participants

The study included PHNs who met the following inclusion criteria: 1) working in maternal and child health or child and family welfare departments; 2) having more than 3 years of experience; and 3) handling child neglect-related abuse cases within the last 3 years. PHNs who were not working at the time of the survey because of maternity leave, childcare leave, sick leave, or similar reasons were excluded. At the department level, a census approach was applied, distributing questionnaires to all maternal and child health departments in the selected prefectures. At the individual level, PHNs were included through purposive sampling based on predefined eligibility criteria.

Approximately 30 % of child welfare workers experienced severe secondary trauma [22], therefore, we predicted a prevalence of depression of 30 %. Thus, the required sample size was determined using the single population proportion formula ($n = Z^2 \times P \times (1-P)/d^2$) [23], with an estimated depression prevalence of 30 % ($P = 0.30$), 95 %CI ($Z = 1.96$), and a 5 % margin of error ($d = 0.05$). As such, a minimum sample size of 323 was needed. In a nationwide survey of local governments on child neglect, the response rate was approximately 20 % [24]. Assuming a 20 % response rate and a 90 % valid response rate, at least 1,795 questionnaires were needed. Finally, 1,800 questionnaires were mailed to 534 maternal and child health departments.

2.3. Study measures

2.3.1. Dependent variables

The Patient Health Questionnaire-2 (PHQ-2) was developed by Kroenke et al. to screen for depression [25]. The questions addressed "having little interest or enjoyment in whatever I try to do" and "feeling heavy, depressed, or hopeless" in the past 2 weeks. They were rated on a 4-point Likert scale — 0 (not at all), 1 (a few days), 2 (more than half the days), and 3 (almost every day) — with a total score ranging from 0 to 6. The PHQ-2 was chosen for its brevity, diagnostic accuracy, and suitability for survey research with minimal response burden. A meta-analysis has demonstrated the PHQ-2's utility as a brief screening tool for depressive symptoms, showing good sensitivity and specificity as a preliminary step before structured diagnostic interviews based on Detect Major Depression or the International Statistical Classification of Diseases criteria [26]. The scale's sensitivity is 0.83, and its specificity is 0.92. In the present study, a score of $\geq$3 was used as the cut-off for depression, with PHQ-2 scores $\geq$3 and < 3, indicating the groups with and without depression, respectively.

2.3.2. Independent variables

Since individual and work environment factors influence workers' mental health [27], these factors were selected as potential influences on the occurrence of depression in this study.

2.3.2.1. Individual factors. Basic attributes and individual factors were ascertained, including years of experience, whether the participants had experience in disability welfare, and whether they had handled severe cases within the past year.

2.3.2.1.1. Years of experience (career stage). The need to consider the impact of years of experience on mental health has been highlighted [28]. In Japan, the career stages of PHNs have traditionally been classified into three categories—new, mid-career, and management—based on years of practical experience. Following recent studies [29], this study classified years of experience as follows: $\leq$5 years, new-stage PHNs; 6–24 years, mid-career-stage PHNs; and $\geq$25 years, management-stage PHNs.

2.3.2.1.2. Experience in disability welfare work. The background of neglect-related abuse cases includes the parent's mental or intellectual disability and the child's disability [30]. Additionally, an understanding of disability characteristics and welfare services is important. PHNs with experience in welfare services may leverage their experience to deal with neglect-related abuse cases, and the presence or absence of experience in welfare services may have an impact on mental health.

2.3.2.1.3. Assignment of high-risk cases within the past year (being in charge of a case that required cooperation with the Child Guidance Centre within the past year). PHNs report an increased risk of burnout and lower mental health-related quality of life as job demands increase [18]. Patients in collaboration with the Child Guidance Centre are at increased risk. Therefore, PHNs require higher skills associated with greater psychological burdens. Respondents were asked if they had been assigned a case requiring collaboration with the Child Guidance Centre within the past year.

2.3.2.1.4. Work controls. Low work control is believed to have a significant impact on depression [31]. Work control was measured using the Individual Public Health Nurse Skills subscale of the Individual and Organizational Readiness Assessment Scale for public health nurses' decision-making to address child neglect [32]. This scale measures PHNs' readiness to make decisions to support families experiencing neglect-related abuse. The scale consists of 15 items with four factors: individual PHNs' values and attitudes, individual PHN skills, perceptions and relationships within the workplace, and perceptions of and relationships with related agencies. Although we used only the subscales and not the total scale score, all subscales were included in the analysis, and the reliability and validity of these subscales have been previously established using the same dataset [32]. The total score on the scale ranged from 15 to 75, with higher scores indicating better readiness to provide support. A Cronbach's α coefficient of 0.871 for reliability and a correlation coefficient of 0.465 for criterion-related validity have been confirmed [32]. We used the research equipment after obtaining permission from the developer. The subscale "individual PHN skills" consists of content related to case management, ensuring response time, and clarifying the basis for decisions. It comprises seven items rated on a 5-point Likert scale (1 = not at all applicable to 5 = very applicable), with total scores ranging from 7 to 35. The internal consistency of this subscale was confirmed by a Cronbach's α coefficient of 0.813. This was adopted because it allows for assessing whether they are in control of their work.

2.3.2.1.5. Psychosomatic controls. The "individual PHNs' values and attitudes" subscale of the Individual and Organizational Readiness Assessment Scale for Public health nurses' decision-making to address child neglect was used to assess whether PHNs could understand and manage their own mental and physical conditions when handling cases [32]. This subscale comprises three items rated on a 5-point Likert scale (1 = not at all applicable to 5 = very applicable), with total scores ranging from 3 to 15. The internal consistency of this subscale was confirmed by a Cronbach's α coefficient of 0.743. Having a way to cope with one's emotional state has been shown to affect the occurrence of compassion fatigue [33]. Thus, the ability of PHNs to grasp and control their own mental and physical states is considered to affect mental health.

2.3.2.2. Workplace environmental factors

2.3.2.2.1. Support from within the workplace. The "perceptions and relationships within the workplace" subscale of the Individual and Organizational Readiness Assessment Scale for public health nurses' decision-making to address child neglect [32] was used to measure perceptions of emotional support from within the workplace and collaboration in the workplace in handling cases. This subscale comprises two items rated on a 5-point Likert scale (1 = not at all applicable to 5 = very applicable), with total scores ranging from 2 to 10. The internal consistency of this subscale was confirmed by a Cronbach's α coefficient of 0.758. Furthermore, the scale assesses whether cases are shared within the workplace and whether emotional support is obtained. Nurses found that workplace support had a reduced impact on depression [34], and the degree of support within the workplace was considered to influence mental health.

2.3.2.2.2. Relationships with related agencies. The "perceptions of and relationships with related agencies" subscale of the Individual and Organizational Readiness Assessment Scale for public health nurses' decision-making to address child neglect [32] was used to measure the perception of collaboration with related agencies in case response and the relationship between PHNs and associated agencies. This subscale comprises three items rated on a 5-point Likert scale (1 = not at all applicable to 5 = very applicable), with total scores ranging from 3 to 15. The internal consistency of this subscale was confirmed by a Cronbach's α coefficient of 0.803. Thus, the scale evaluates whether the nurses have a good relationship with associated agencies. The difficulty of cooperation and collaboration with related agencies has been cited as a challenge in case response [35]. Therefore, a good PHN relationship with related agencies may affect their mental health.

2.3.2.2.3. Regular meetings in the workplace. Frequent communication with superiors and colleagues has an impact on depression [36]. In the present study, we inquired about the frequency of

meetings held primarily for case management, as well as the frequency of communication. Respondents were divided into two categories: infrequent (not at all, once a year, once every few months, or once a month) and frequent (once a week or when necessary).

2.3.2.2.4. On-the-job training experience in the workplace. Child welfare professionals with less in-service education were significantly more likely to experience occupational stress [37]. This study investigated whether PHNs had on-the-job training experience in the workplace.

2.3.2.2.5. Perceptions of manpower availability. Evidence suggests that subjective workload is related to mental health [28]. The present study measured workload by asking respondents whether they were well-staffed.

2.4. Data collection

The Child and Family Welfare Department and the Maternal and Child Health Department were asked to administer the questionnaires to PHNs within their respective departments. The purpose, objectives, and methods of the study were explained in writing to the department heads, who verified the eligibility of the PHNs and distributed the self-administered anonymous questionnaires. Participants returned the completed questionnaires individually by mail. To improve the response rate, the survey form was kept concise. To identify eligible participants, the questionnaire included screening questions at the beginning, asking whether respondents had more than 3 years of experience in maternal and child health or child and family welfare and whether they had handled child neglect-related abuse cases within the past 3 years. A total of 372 respondents participated in this study (response rate: 20.6 %). Of these, 13 were excluded because of missing data, resulting in a final sample size of 359 (valid response rate: 96.5 %).

2.5. Data analysis

Descriptive statistics were calculated for all variables. The χ^2 and Mann–Whitney *U* tests were performed to examine the association between depression and each of the 10 items: five items for individual factors and five for environmental factors. Multivariate logistic regression analysis was conducted, with or without depression as the dependent variable and 10 items as independent variables. Before logistic regression analysis, the absence of multicollinearity among the explanatory variables was confirmed by a correlation coefficient ($r < 0.7$) and a variance inflation factor <10. Additionally, to compare the effects of independent variables on different scales, all continuous independent variables were standardized (transformed so that the mean was zero and the standard deviation was 1). For categorical variables, the reference categories were set according to analytical intent and interpretability. For years of experience (career stage), “new-stage PHNs” were used as the reference category, with comparisons made to “mid-career-stage” and “management-stage” PHNs. For binary variables, the reference category was set as “Yes” or “No” depending on interpretability. Specifically, for experience in disability welfare work, “Yes” was the reference, and for assignment of high-risk cases within the past year, “No” was the reference. Variables were selected using the forced entry method. Statistical significance was set at $P < 0.05$. The Hosmer–Lemeshow test was used to assess the model’s goodness of fit. Statistical analyses were conducted using IBM SPSS version 29.

2.6. Ethical considerations

This study was approved by the Ethics Committee of Kumamoto University (No. 2448) on April 1, 2020. As this study was a secondary analysis using the same dataset as the previously published scale development study, it was covered by the same approved research protocol. The secondary analysis was conducted under this existing ethical approval. After explaining the purpose and content of the study and ethical considerations to the research participants in a letter requesting participation in the survey, they responded to the survey of their own free will, and their consent was deemed to have been obtained upon the return of the survey form.

3. Results

3.1. Overview of study participants

Regarding years of experience, 75 (20.9 %), 207 (57.7 %), and 77 (21.4 %) participants were new-stage, mid-career-stage, and management-stage PHNs, respectively. The mean PHQ-2 score was 1.12 (standard deviation: 1.48). A total of 41 (11.4 %) and 318 (88.6 %) participants were with and without depression, respectively. 127 (35.4 %) participants had a PHQ-2 score ≥2, although below the standard cut-off for depression, which has been interpreted in some studies as indicating moderate depressive symptoms. The detailed information of all the participants is shown in Appendix A.

3.2. Related factors of depression among public health nurses engaged in child neglect-related abuse cases

3.2.1. Univariate analysis

3.2.1.1. Individual factors. Among those with depression, 9.8 % were new-stage, 53.7 % were mid-career, and 36.6 % were management-stage PHNs. The depression group included a markedly higher proportion of management-stage PHNs compared with new-stage PHNs, there is a statistically significant difference between them ($\chi^2 = 7.82$, $P = 0.020$). Regarding experience with disability welfare work, the percentage of participants without depression was higher among those with such experience ($\chi^2 = 5.61$, $P = 0.018$). The scores for work controls were significantly lower for those with depression ($Z = -3.941$, $P < 0.001$). However, no significant differences were observed between the two groups regarding their responsibility for cases requiring cooperation with the Child Guidance Centre ($\chi^2 = 3.78$, $P = 0.052$) and psychosomatic controls ($Z = -1.163$, $P = 0.245$) (Appendix A).

3.2.1.2. Workplace environmental factors. The scores for perceptions and relationships within the workplace (i.e., the status of support within the workplace) were significantly lower for those with depression ($Z = -5.287$, $P < 0.001$). The percentage of those who held regular meetings as needed or as often as once a week (i.e., frequent) was significantly lower among those with depression ($\chi^2 = 4.39$, $P = 0.036$). Regarding perceptions of manpower availability, the proportion of respondents with probable depression was significantly higher among those who reported that manpower was unavailable ($\chi^2 = 11.98$, $P < 0.001$). PHNs with depression tended to have lower scores for perceptions of and relationships with related agencies compared with those without depression, despite similar median scores between the two groups ($Z = -2.399$, $P = 0.016$). No significant differences were observed between the two groups in terms of on-the-job training experience in the workplace ($\chi^2 = 0.02$, $P = 0.897$) (Appendix A).

3.2.2. Multivariate logistic regression analysis

3.2.2.1. Individual factors. Being a management-stage PHN, compared with a new-stage PHN, was significantly associated with depression ($OR = 6.27$, 95 %*CI*: 1.63–24.18, $P < 0.010$), whereas mid-career stage PHN was not ($OR = 1.74$, 95 %*CI*: 0.50–6.05, $P = 0.385$). In addition, without experience in disability welfare work ($OR = 3.49$, 95 %*CI*: 1.05–11.55, $P = 0.041$), being in charge of a case that required cooperation with the Child Guidance Centre within the past year ($OR = 8.28$, 95 %*CI*: 2.07–33.08, $P < 0.010$), and lower work controls ($OR = 0.53$, 95 %*CI*: 0.32–0.86, $P < 0.010$) were associated with depression (Table 1).

3.2.2.2. Workplace environmental factors. Higher scores for perceptions and relationships within the workplace were significantly associated with a lower likelihood of depression ($OR = 0.43$, 95 % *CI*: 0.29–0.65, $P < 0.001$). In contrast, perceiving manpower as not secured, compared with secured, was significantly associated with a likelihood of depression ($OR = 9.07$, 95 %*CI*: 1.17–70.40, $P = 0.035$) (Table 1). The model's goodness of fit was 89.7 % for discriminant accuracy (Hosmer–Lemeshow test, $P = 0.792$; Nagelkerke $R^2 = 4.668$).

4. Discussion

4.1. Depression status among the public health nurses

In this study, 41 (11.4 %) respondents were possibly depressed. In a study examining depression among healthcare workers, the prevalence of depression was 15 % among managers [38]. Dealing with child abuse is considered part of health crisis management, alongside disaster and pandemic response [39]. Comparable depression rates have been reported among public health professionals during crises, for example, 15 % prevalence among public servants during the Great East Japan Earthquake [40]. In addition, 35.4 % of participants in this study had a PHQ-2 score ≥2. Although this does not meet the standard cutoff for depression, some studies interpret PHQ-2 scores ≥2 as indicative of moderate depressive symptoms [41], and this may suggest the presence of a group at risk of future deterioration if unaddressed. Child welfare professionals experiencing high psychological burden, including secondary traumatic stress, have been shown to exhibit reduced work performance and unhealthy behaviours [13]. These findings raise the concern that psychological distress among PHNs working in maternal and child health and child welfare may compromise the quality of support services. Therefore, this study highlights the urgent need to promote mental health among PHNs in these fields. The results also suggest the importance of implementing organizational measures to prevent depression and support sustained service quality in PHN practice.

4.2. Individual factors related to depression

Being in charge of a case that required cooperation with a Child Guidance Centre within the past year was significantly associated with depression in the multivariate analysis. The increase in complex and difficult individual consultations and urgent response cases was cited as a burden for PHNs [42]. PHNs reported that their support did not proceed as expected; uncertainty and anxiety about the threat to the child's life increased, and they felt pressured and isolated, as though they were left alone to make important decisions and respond [43]. PHNs dealing with urgent mental health cases are reported to have high levels of stress [44]. As described above, the mental burden of encountering high-risk cases has been highlighted in previous studies, and this study confirmed that the mental health of PHNs was affected. Handling cases with complex backgrounds and difficult responses requires a high level of competence and imposes a substantial mental burden. Similar to child welfare workers, professionals may face a high risk of secondary traumatic events [12]. Of the respondents in the present study, 81.6 % had been assigned a case that required collaboration with a Child Guidance Centre within the past year. In a previous survey, 70 % of PHNs had experience collaborating with the Child Guidance Centre [45]; that percentage was even higher in this survey. The number of PHNs dealing with serious cases may be increasing. It can be assumed that many PHNs are frequently assigned high-risk cases, highlighting the importance of providing mental healthcare for all PHNs managing high-risk cases.

Regarding years of experience, 60 % of the respondents in this study were mid-career-stage PHNs, and most PHNs providing child neglect-related abuse support were also in this stage. In the multivariate analysis, being a management-stage PHN, compared with a new-stage PHN, was significantly associated with

Table 1
Logistic regression analysis of factors related to depression.

Factors	Univariate model (Non-standardised)			Multivariate model (Standardised)				
	OR	95 %*CI*	*P*	*β*	*SE*	*OR*	95 %*CI*	*P*
Individual factors								
Years of experience (Career stage) (Ref.: New-stage)								
Mid-career stage	2.11	0.70–6.34	0.183	0.55	0.64	1.74	0.50–6.05	0.385
Management stage	4.29	1.35–13.62	0.013	1.84	0.69	6.27	1.63–24.18	<0.010
Experience in disability welfare (Ref.: Yes)								
No	3.37	1.17–9.75	0.064	1.25	0.61	3.49	1.05–11.55	0.041
Assignment of high-risk cases within the past year* (Ref.: No)								
Yes	3.13	0.94–10.47	0.064	2.11	0.71	8.28	2.07–33.08	<0.010
Psychosomatic controls (Per 1 point increase)	0.90	0.76–1.06	0.203	0.11	0.23	1.12	0.70–1.77	0.634
Work controls (Per 1 point increase)	0.84	0.77–0.92	<0.001	−0.64	0.25	0.53	0.32–0.86	<0.010
Workplace environmental factors								
Perceptions and relationships within the workplace (Per 1 point increase)	0.56	0.45–0.70	<0.001	−0.84	0.21	0.43	0.29–0.65	<0.001
Perceptions of and relationships with related agencies (Per 1 point increase)	0.86	0.71–1.03	0.095	0.31	0.26	1.37	0.83–2.26	0.226
Regular meetings in the workplace (Ref.: Frequent)								
Infrequent	2.07	1.04–4.15	0.039	0.32	0.21	1.38	0.92–2.06	0.117
On-the-job training experience in the workplace (Ref.: Yes)								
No	1.04	0.54–2.00	0.898	−0.29	0.40	0.75	0.34–1.63	0.472
Perceptions of manpower availability (Ref.: Secured)								
Not secured	14.80	2.01–109.53	<0.010	2.20	1.04	9.07	1.17–70.40	0.035

Note: * Being in charge of a case that required cooperation with the Child Guidance Centre within the past year.

depression, whereas the mid-career stage was not. Previous studies have also reported that managers tend to experience higher levels of occupational stress [46]. Moreover, positions that require more overtime and holiday work are associated with a higher percentage of managers [47]. Furthermore, research has shown that the workload increases for mid-management PHNs [48] owing to the inexperience of new-stage PHNs. Those with more years of experience tended to have higher workloads and may have been in a high-stress state. Child neglect-related abuse is handled by organizations [24], and managers may be responsible for supporting decisions and providing explanations, which may place a mental burden on them. Workload reduction measures, such as adjusting the workload within the workplace and not working on off days, are important considerations.

Moreover, the proportion of respondents without depression was significantly higher among those with experience in disability welfare work. The background of child neglect-related abuse includes overlapping factors, such as parents' mental and psychological problems, domestic violence, children's mental and physical problems, and poverty [49]. Challenges in handling neglect-related abuse cases include intervention methods, support systems [8] PHNs' experiences in welfare work have led to a significant change in the quality perspective from which they view life, the practice of creating social resources [50], consideration of physical and emotional support needs, and the ability to develop a deep understanding of cases and support systems, such as securing support from neighbourhoods and related professions [51]. In fiscal year 2013, with the implementation of the Law for Comprehensive Support for Persons with Disabilities, municipalities were centralised in providing services, and the decentralized assignment of PHNs to disability welfare work was promoted. In the same fiscal year, 5.7 % of PHNs in municipalities were assigned to the welfare sector [3]. Among the participants, 24.8 % had experience in disability welfare. Although the majority had no experience, those with experience accounted for a quarter of the total, suggesting that this number is increasing. Most transferred to the welfare sector after 10 years of PHN experience [50]. Performing job rotations in each workplace in a planned manner was considered necessary to enable all PHNs to gain and share their experiences in welfare work.

Regarding work control, nurses' depression is most strongly associated with low work control [36]. Thus, those with greater work control tended to have better mental health. In the present study, higher work control scores were interpreted as reflecting PHNs' ability to manage and evaluate their own work tasks, contributing to a sense of confidence and security. In child neglect-related abuse support, it is necessary for PHNs to confirm whether they can manage their work. We believe the Individual and Organizational Readiness Assessment Scale for Public Health Nurses' Decision-making to address child neglect can be used [32]. This is important for the mental healthcare of the PHNs and for ensuring satisfactory support for the target children. In addition, providing support to enable workers to control their work in the workplace has been shown to positively impact their ability to carry out their duties [52]. Information regarding the status of the cases managed by each PHN can be used for supervision, risk management, and mental healthcare in the workplace.

These findings highlight the importance of addressing individual factors such as case complexity, career stage, welfare experience, and work control in supporting PHNs' mental health. Organizational efforts to provide mental health care, implement structured job rotations, and enhance work autonomy are essential in child protection services.

4.3. Workplace environmental factors related to depression

Workplace environmental factors associated with the likelihood of depression were related to perceptions and relationships within the workplace and perceptions of manpower availability in the logistic regression analysis. Those without depression had higher scores for perception of manpower availability, perceptions and relationships within the workplace, and more support. This study found no association between perceptions and relationships within related agencies, indicating that support within the workplace is more important for the mental health of PHNs. Although not statistically significant in the logistic regression analysis, meeting frequency was also examined as a supplementary factor. Approximately half of those who held regular meetings (47.1 %) had these meetings once a week or when necessary, while half held the meetings infrequently and did not communicate well. The chi-square test showed that the group not likely to be depressed had a significantly higher percentage of those who held meetings weekly or when needed. Frequent discussions were considered part of the workplace culture for easy communication. Communication and teamwork in the workplace can influence the emotional control of the supporters [33]. Acceptance and cooperation within the workplace are important for enhancing the emotional aspects of supporters [53], and we believe that discussing concerns immediately and sharing cases within the workplace without holding them in will reduce the emotional burden. When dealing with child abuse, burnout can be reduced by maintaining hope in achieving one's goals [54]. PHNs need to support each other to maintain hope when dealing with cases within the workplace.

In contrast, perceptions of manpower availability were significantly associated with depression. A significantly higher percentage of respondents who reported insufficient manpower availability were more likely to be depressed. The adjusted OR of 9.07 highlights the seriousness of this factor. The lack of manpower means that the workload is heavy, there are few counsellors, and the physical and psychological burdens are high. High workloads due to understaffing can affect burnout and case handling among child welfare workers [55], and the same risk may apply to PHNs. In a previous study, a significantly higher proportion of respondents in the depressed group reported that they could not handle their workload in time compared with those in the non-depressed group [56]. High workloads are associated with stress responses in both the minds and bodies of PHNs [48]. The more overtime PHNs work and the greater their overall workload, the lower their self-assessment of work-life balance [57]. PHNs tend to place importance on work-life balance [58], so they are likely to experience psychological conflict when their workload is heavy. PHNs who intend to leave their jobs often have higher stress levels [59], which may lead to staffing shortages. A workplace environment that is easy to consult and obtain support can be expected to buffer the impact on mental health; however, obtaining support may be difficult in understaffed environments. Ensuring sufficient staffing is thus vital for safeguarding PHNs' mental health and sustaining child welfare services.

These findings suggest that fostering a supportive workplace culture and ensuring adequate staffing are critical strategies for promoting the mental health of PHNs and maintaining stable, high-quality child health and welfare services.

4.4. Limitations

As this was a cross-sectional study, a causal relationship could

not be established. Therefore, a longitudinal study is required to elucidate this causal relationship better. In addition, because this study focused on occupational duties, other factors, such as participants' personal backgrounds and personality traits, were not fully explored. Additionally, using a two-item measure to assess symptoms of depression may have limitations in sensitivity or specificity compared to a longer measure, such as the PHQ-9. However, for the study, it was important to consider the data collection burden in this population. Additionally, PHQ-2 is a screening tool for depression. This study only conducted a preliminary exploration. It is suggested that future research should further employ diagnostic tools for depression for evaluation in order to comprehensively explore the depression situation of this population. Without prior knowledge of the number of PHNs in each maternal and child health department, the questionnaires were uniformly distributed to four individuals each, which may have resulted in over- or under sampling. The return rate was low, which may have biased the sample toward responses from a particular group of PHNs. The fact that we left it up to the recipients to verify participants' eligibility and distribute the questionnaires may have affected the quality control of the survey responses.

5. Conclusions

In this study, 81.6 % of PHNs engaged in child neglect-related abuse cases were responsible for high-risk cases. In addition, 11.4 % of participants were possibly depressed. Logistic regression analysis revealed that factors associated with depression included years of experience, experience in disability welfare work, being responsible for a case requiring cooperation with a Child Guidance Centre within the past year, and work controls. Workplace environmental factors were related to perceptions and relationships within the workplace and perceptions of manpower availability. Implications for mental healthcare for PHNs providing child neglect-related abuse support include managing their workload effectively. In addition, the following are recommended for workplaces: securing sufficient manpower, reducing physical and mental burdens, considering both quantitative and qualitative burdens of management-stage PHNs, fostering an environment of frequent communication and mutual support, and providing opportunities for all PHNs to gain experience in disability welfare work.

Data availability statement

The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Ayano Ogata: Methodology, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review & editing, and Project administration. **Ayako Ide-Okochi:** Conceptualization, Methodology, Validation, Formal analysis, Writing - original draft, Writing - review & editing, and Supervision.

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Declaration of competing interest

The authors of this study have declared no conflict of interest.

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Appendices. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijnss.2025.08.006.

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